Assignment 1 - DML / DDL Practice with Solutions

DML Section

- Insert a new employee into the `employees` table.

```
INSERT INTO employees (employeeID, firstName, lastName, department, salary) VALUES (201, 'John', 'Doe', 'IT', 45000);
```

- Update the salary of an employee whose ID is 105 to 50,000.

```
UPDATE employees SET salary = 50000 WHERE employeeID = 105;
```

- Delete all employees from the `employees` table who have a salary less than 10,000.

```
DELETE FROM employees WHERE salary < 10000;
```

- Increase salary by 15% for all employees in the 'Marketing' department.

```
UPDATE employees SET salary = salary * 1.15 WHERE department = 'Marketing';
```

- Copy all rows from `products` to a new table called `products_backup`.

```
CREATE TABLE products_backup AS SELECT * FROM products;
```

- Set all `shippedDate` values to today's date for orders that are not yet shipped.

```
UPDATE orders SET shippedDate = CURRENT_DATE WHERE shippedDate IS NULL;
```

- Change the department name from 'Sale' to 'Sales' in the `employees` table.

```
UPDATE employees SET department = 'Sales' WHERE department = 'Sale';
```

- Insert a new product into the `products` table with appropriate dummy data.

```
INSERT INTO products (productCode, productName, productLine, buyPrice) VALUES ('P101', 'Test Product', 'Classic Cars', 55.00);
```

- Delete all orders with NULL as `orderDate`.

```
DELETE FROM orders WHERE orderDate IS NULL;
```

- Update all `customerNames` to uppercase in the `customers` table.

```
UPDATE customers SET customerName = UPPER(customerName);
```

DDL Section

- Create a new table called `departments` with fields: `dept_id`, `dept_name`, and `location`.

```
CREATE TABLE departments (dept_id INT PRIMARY KEY, dept_name VARCHAR(50), location VARCHAR(100));
```

- Add a new column `bonus` of type `DECIMAL(10,2)` to the `employees` table.

```
ALTER TABLE employees ADD bonus DECIMAL(10,2);
```

- Rename the `products_backup` table to `archived_products`.

  RENAME TABLE products_backup TO archived_products;

- Drop the `temp_sales` table if it exists.

  DROP TABLE IF EXISTS temp_sales;

- Modify the `salary` column in the `employees` table to allow NULL values.

  ALTER TABLE employees MODIFY salary DECIMAL(10,2) NULL;

- Add a primary key constraint to the `departments` table on `dept_id`.

  ALTER TABLE departments ADD CONSTRAINT pk_dept PRIMARY KEY (dept_id);

- Create a view called `high_earners` that shows employees with salary above 60,000.

  CREATE VIEW high_earners AS SELECT * FROM employees WHERE salary > 60000;

- Create a new index on the `lastName` column in the `employees` table.

  CREATE INDEX idx_lastname ON employees(lastName);

- Add a unique constraint to the `email` column in the `employees` table.

  ALTER TABLE employees ADD CONSTRAINT unique_email UNIQUE (email);

- Create a new table `audit_log` with columns: `action_id`, `action_type`, `timestamp`.

  CREATE TABLE audit_log (action_id INT PRIMARY KEY, action_type VARCHAR(50), timestamp DATETIME);